Supplemental Materials

Leps-P: Probabilistic Strength Envelopes with Stress-Decaying Curvature for Rockfill and Mine Waste Rock

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This file contains Figs. S1–S5 and Tables S1–S2, cited in the paper. Equation and table numbers without the prefix S refer to the paper.

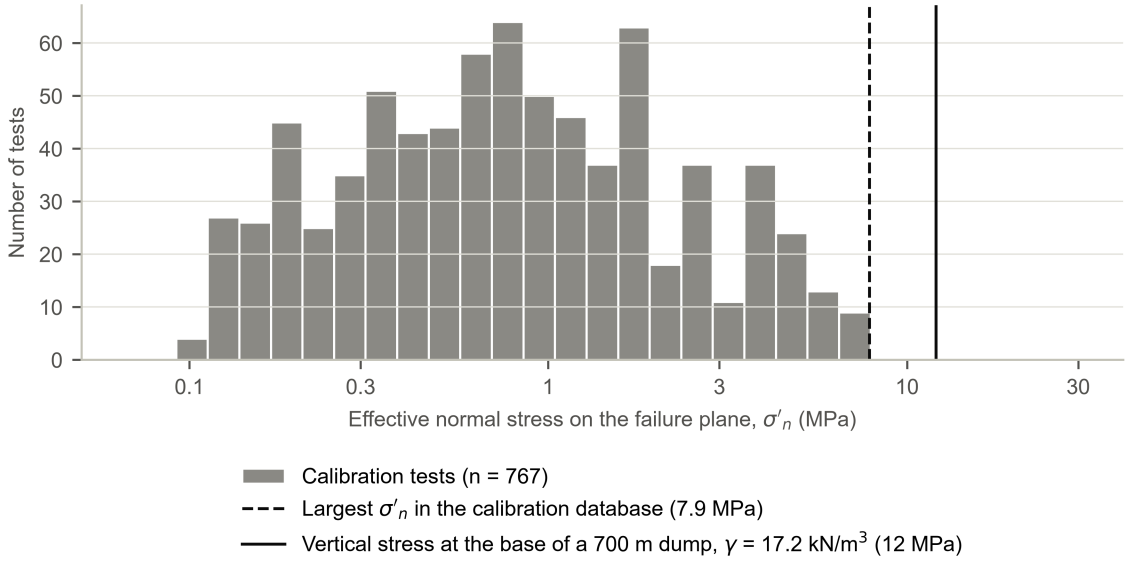


Fig. S1. Distribution of effective normal stress on the failure plane in the calibration database.

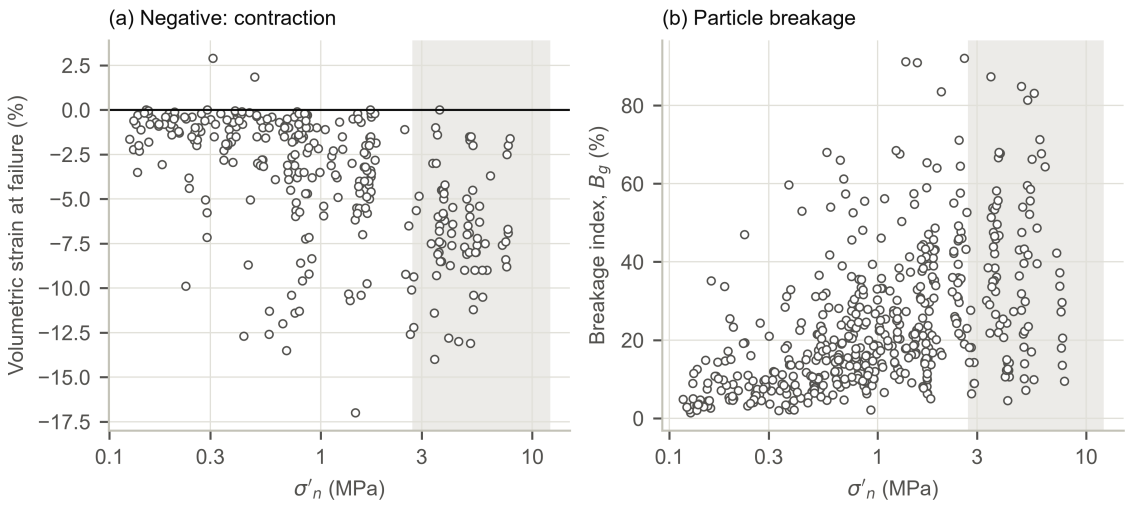


Fig. S2. Net volumetric strain at the maximum deviator (negative: contraction) and breakage index of Marsal (1973) versus σ'_n in the calibration database; shaded: $\sigma'_n > 2.75$ MPa. Reported breakage indices above 100%, which are impossible by definition, were excluded (4 tests). Above 2.75 MPa the residuals of Leps-P were unrelated to the breakage index (Spearman $\rho = 0.00$, n = 91).

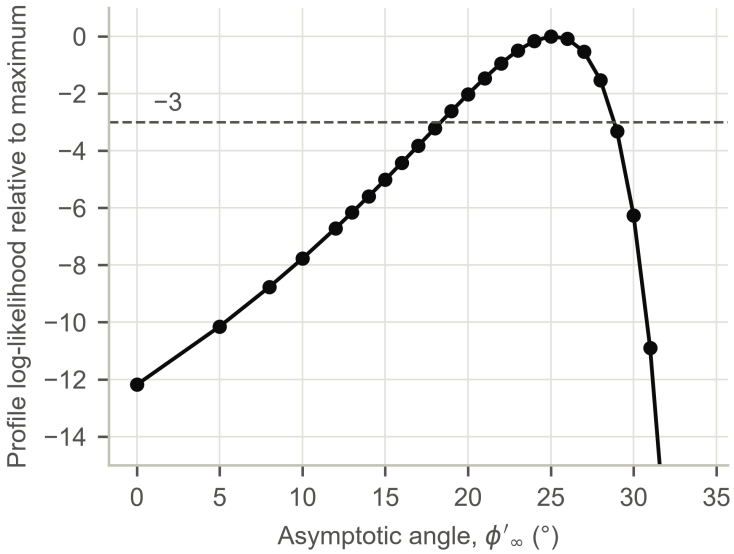


Fig. S3. Profile log-likelihood of the asymptotic angle ϕ'_∞ relative to its maximum (1685.51); at each angle the population decay rate ζ_0 and the variance parameters were re-estimated by maximum likelihood. $\phi'_\infty = 0$ corresponds to the form without an asymptote. Dashed: 3 log-likelihood units below the maximum.

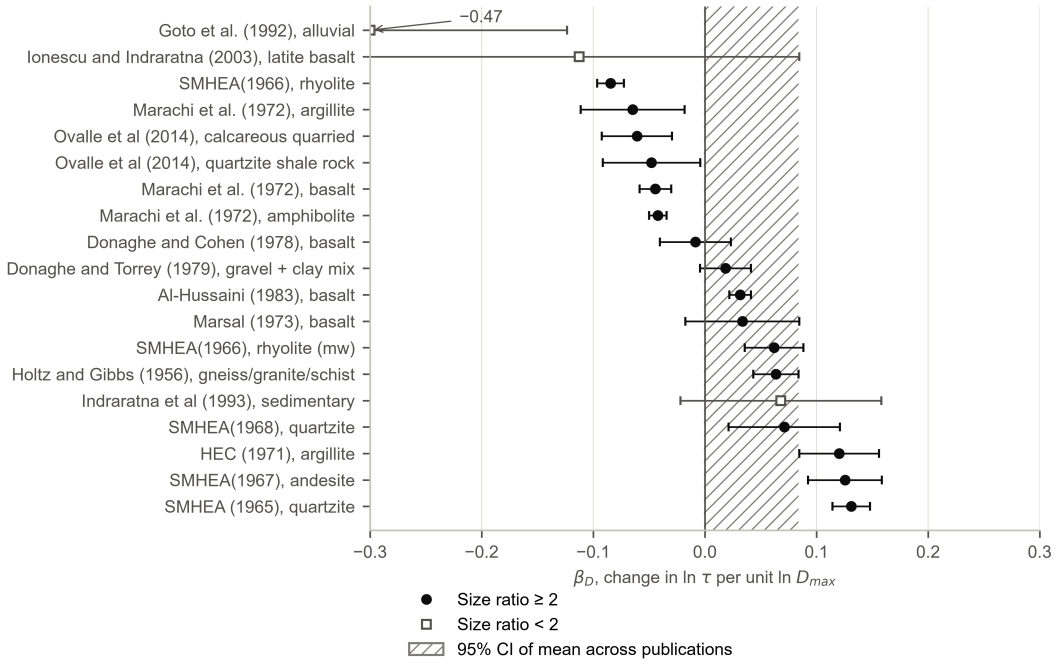


Fig. S4. Within-material specimen-size coefficient β_D (bars: 95% confidence intervals; set intercepts adjusted for void ratio). Hatched band: Knapp–Hartung 95% confidence interval of the random-effects mean across 12 publications with size ratio ≥ 2 (+0.041; −0.001 to +0.084). One value outside the axis is annotated.

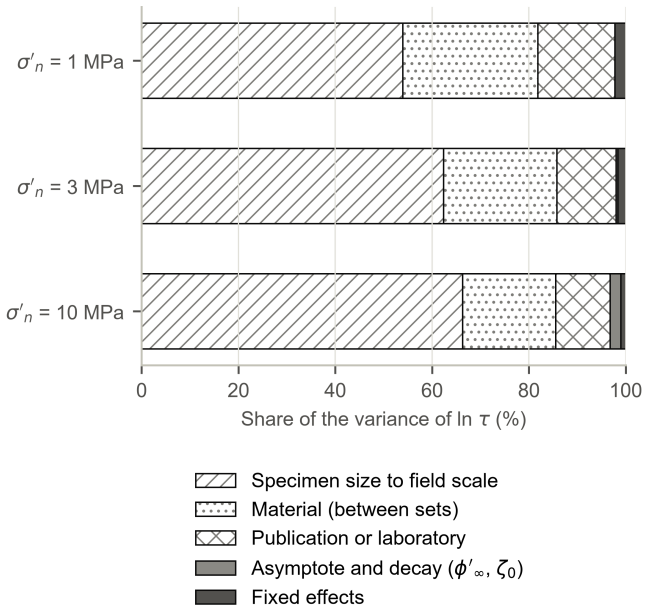


Fig. S5. Shares of the variance of the mean-envelope ln τ for an untested material of low rock quality placed at e = 0.65, scaled from 0.18 m to 1 m maximum particle size.

TABLE S1. Mode B: every procedure evaluated (held-out tests at $\sigma'_n \geq 2$ MPa predicted from tests at $\sigma'_n \leq 1.5$ MPa of the same set)

Procedure	80% coverage	Below P10	SD of z	CRPS	Δ vs level update (95% CI)	Holm-adjusted p
Prior, no site tests	95%	3%	0.76	0.0424	+0.0115 (+0.0043 to +0.0179)	0.48
Least-squares fit of ϕ'_1, m = 0.90	81%	17%	0.86	0.0407	+0.0098 (−0.0011 to +0.0165)	1.00
Least-squares fit of ϕ'_1, m = median within-set slope of training data	81%	2%	1.00	0.0369	+0.0061 (−0.0021 to +0.0207)	0.74
Joint update of both random effects	80%	12%	1.34	0.0330	+0.0022 (−0.0056 to +0.0200)	1.00
Level update of each set (Eq. 8)	83%	9%	0.91	0.0308	—	—
Level update after discarding flagged sets	83%	12%	0.97	0.0345	+0.0036 (−0.0041 to +0.0106)	1.00
Recommended procedure: discard flagged sets, joint update with shared laboratory effect	83%	12%	0.98	0.0344	+0.0036 (−0.0044 to +0.0104)	1.00

Note: 64 tests, 46 sets, 13 publications. The procedures were developed while examining earlier versions of the model on the same held-out sets; the comparisons are exploratory. p-values from exact sign-flip tests on publication means, Holm-adjusted within Mode B. CRPS in units of ln τ.

TABLE S2. Secant friction angle ϕ'_s (°) of the waste rock of the example, median / 5th percentile of the mean envelope

Material	Laboratory, 1 MPa	Placed, 1 MPa	Placed, 2 MPa
Rock type A	40.5 / 39.1	35.3 / 30.3	33.4 / 28.6
Rock type B	43.1 / 38.6	38.2 / 31.5	35.9 / 29.8
Rock type C	41.7 / 40.3	37.2 / 32.0	35.2 / 30.2
Rock type D (untested)	40.9 / 36.6	36.1 / 29.5	34.2 / 28.0
Rock type E (untested)	40.9 / 36.5	33.9 / 27.4	32.3 / 26.3
Dump average, model correlation	41.5 / 39.7	36.4 / 31.1	34.4 / 29.4
Dump average, full correlation	41.5 / 38.7	36.4 / 30.6	34.4 / 29.0

Note: Laboratory: specimen void ratio and size. Placed: void ratio 0.65 (rock type E, 0.80), maximum particle size 1 m, size effect with zero mean. The full-correlation case sorts the samples of each rock type at every stress, an upper bound on dependence. No site tests above $\sigma'_n \approx 4.6$ MPa. The critical surfaces of Table 4 of the paper mobilize σ'_n of up to about 2 MPa.